

Process Characterization Plan

A-Mab Drug Substance — Production Bioreactor (Step 3) — PCP-003

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ANOVA, analysis of variance · CCD, central-composite design · CPP, critical process parameter · CQA, critical quality attribute · DoE, design of experiments · DS, drug substance · GPP, general process parameter · HCP, host-cell protein · ICH, International Council for Harmonisation · KPP, key process parameter · MSAT, Manufacturing Science and Technology · NOR, normal operating range · PAR, proven acceptable range · PPQ, process performance qualification · QbD, Quality by Design · RRF, risk ranking and filtering · RSM, response-surface methodology · SDM, scale-down model · SEC, size-exclusion chromatography · VCD, viable cell density · WC-CPP, well-controlled CPP.

1 Purpose and scope

This Process Characterization Plan (**PCP-003**) defines the Stage 1 (Process Design) characterization of the A-Mab production bioreactor (Step 3), the fed-batch cell-culture operation that establishes the N-glycosylation, charge-variant and initial size-variant and process-impurity profile of the drug substance. The plan specifies the objectives, the quality attributes in scope, the process parameters and ranges to be characterized, the experimental designs, the analytical methods, the sampling plan, the statistical-analysis approach, and the acceptance and decision criteria by which each parameter will be classified and a design space established.

Scope. The 15,000-L commercial-scale production bioreactor, characterized on a qualified 2 L scale-down model (SDM). Seed-train expansion (Steps 1–2) and harvest (Step 4) are addressed only where they affect the production culture; their characterization is out of scope. This plan governs execution only — the results are reported in the paired Process Characterization Report (**PCR-003**).

Basis. The unit-operation model, quality-attribute framework and risk methodology follow the A-Mab case study (CMC Biotech Working Group 2009); the Stage 1 approach and report structure follow PDA TR 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023); the QbD framework follows ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012) and the lifecycle approach of the FDA 2011 guidance (U.S. Food and Drug Administration 2011).

2 Objectives

The characterization will:

1. establish the quantitative relationships between the production-bioreactor process parameters and the cell-culture CQAs;
2. identify the significant main effects, two-factor interactions and curvature;
3. define the multivariate design space over which all CQAs are met simultaneously;
4. establish proven acceptable ranges (PARs) and confirm normal operating ranges (NORs) for each parameter;
5. classify each parameter for its risk of impact to CQAs and to process performance (CPP / WC-CPP / KPP / GPP); and
6. provide the process understanding and ranges that justify the Stage 2 operating conditions and the drug-substance control strategy.

3 Related documents

This plan is executed under the master documents below and is paired with its report; the full package is cross-referenced for traceability.

Table 3.1: Related documents in the A-Mab process-characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-003	Process Characterization Report (Production Bioreactor (Step 3))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The governing procedures and validated analytical methods referenced by this plan are listed in Table 3.2 (numbers are placeholders in this synthetic set).

Table 3.2: Referenced standard operating procedures and analytical method validations.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

Reference	Title	Type
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4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The production bioreactor is a fed-batch mammalian cell culture operated at 15,000 L working volume. It is the only upstream operation that forms product-quality CQAs and is therefore the sole upstream step used to define the upstream design space (CMC Biotech Working Group 2009). Platform knowledge from related humanized IgG1 antibodies and A-Mab development experience established that Fc glycosylation governs ADCC/CDC potency and is set in culture, that aggregate formed in culture is polished downstream, and that the harvest HCP and DNA burden is generated upstream and cleared by the purification train. This knowledge frames the CQAs in scope and the prioritization of parameters for characterization.

4.2 Quality attributes in scope

The characterization targets the CQAs set or principally controlled by the production bioreactor (Table 4.1). Criticality is scored on the A-Mab continuum using Tool #1 (Risk Score = Impact $\times$ Uncertainty); attributes rated High (H) or Very High (VH) are treated as *Critical* (CMC Biotech Working Group 2009; International Council for Harmonisation 2023).

Table 4.1: CQAs in scope, with acceptance criteria and criticality (Tool #1 = Impact $\times$ Uncertainty).

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2–13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3–10 % of glycans	VH	80
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6

4.3 Risk-based prioritization of parameters

Consistent with the A-Mab lifecycle of iterative risk assessments and the risk flow of PDA TR 60, the study scope is set by the Pre-Characterization Process Risk Assessment (**RA-001**). A risk-ranking-and-filtering (RRF) screen scores each parameter for its potential main-effect and interaction impact on the CQAs and assigns the study type accordingly: parameters with a credible multivariate risk are studied in the multivariate DoE; parameters expected to affect only process performance are studied univariately or held at their set-point; parameters with no credible impact over a wide range are not formally studied (CMC Biotech Working Group 2009). The resulting parameter set and study-type assignment are given in Table [6.1](#).

5 Materials and methods

5.1 Scale-down model and its qualification

Characterization will be performed on a 2 L bench-scale bioreactor operated at the commercial working-volume equivalent for pH, temperature, dissolved oxygen and CO₂ control, gas sparging, agitation power per unit volume and feeding strategy (**SOP-2003**, **SOP-2004**). Prior to characterization the SDM will be qualified against at-scale reference data (**SOP-1001**) by comparing the operational parameters and, critically, the input and output quality attributes — viable cell density, viability, titer and the glycan, charge-variant, aggregate, HCP and DNA profiles — to confirm the model is representative and suitable to support design-space and PAR claims (Parenteral Drug Association 2013). The center-point replicates of each design provide an in-study estimate of model reproducibility (pure error) used in the lack-of-fit assessment.

5.2 Cell line, media and culture operation

The production culture is a fed-batch process inoculated from the N-1 seed bioreactor at the target initial viable cell concentration, using platform basal medium and a bolus nutrient feed (**SOP-2004**). Culture pH, temperature, dissolved oxygen and dissolved CO₂ are controlled to set-point; osmolality is a monitored attribute governed by feed and base addition; the culture is harvested on the nominal duration criterion. Raw materials and media are controlled to platform specifications and are treated as identity/quality-controlled inputs rather than characterized variables (CMC Biotech Working Group 2009).

5.3 Analytical methods

CQAs will be measured by the validated methods in Table 5.1; method performance characteristics are per the cited validation reports.

Table 5.1: Validated analytical methods for the characterization.

Reference	Title	Type
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

5.4 Sampling plan

Each DoE run will be sampled at harvest for the released CQAs (glycan map, charge variants, aggregate) and for the harvest HCP and residual-DNA load; daily process samples (viable cell density, viability, metabolites, pH/pCO₂ verification) will confirm the culture trajectory. Center-point runs are replicated within each design (three in screening, four in the response-surface design) to quantify reproducibility and pure error.

5.5 Statistical methods

Designs will be constructed and analyzed in coded factors, with each parameter's characterization range mapped so that the set-point is 0 and the range edges are ± 1 (**SOP-1002**). Screening data will be analyzed by ordinary-least-squares regression of each response on the main effects and all two-factor interactions; factor effects are reported as twice the coded coefficient. Response-surface data will be analyzed with the full quadratic model (main effects, two-factor interactions and pure-quadratic terms). Significance will be assessed at $\alpha = 0.05$. Model adequacy will be judged by R^2 , adjusted R^2 , predicted R^2 (from the PRESS statistic), the overall model F-test, and an ANOVA lack-of-fit test against the center-point pure error. The design space will be defined as the multivariate region over which all CQA responses satisfy their acceptance criteria simultaneously, and commercial-scale capability will be estimated by Monte-Carlo simulation with each parameter varied within its NOR.

6 Study design

6.1 Factors, ranges and study type

Nine parameters are in scope (Table 6.1). The characterization range of each factor (“Range studied”) defines the edges of the explored knowledge space and the basis of the PAR; the NOR is the tighter routine-control range. Classification is an output of the study and is reported in PCR-003. The DoE factor coding is given in Table 6.2.

Table 6.1: Production-bioreactor parameters, set-points, characterization ranges and planned study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Culture pH	pH	6.85	6.6–7.1	6.75–6.95	multivariate
Culture temperature	°C	35	34–36	34.5–35.5	multivariate
Dissolved CO ₂ (pCO ₂)	mmHg	100	40–160	70–130	multivariate
Osmolality	mOsm	400	360–440	375–425	multivariate
Culture duration	days	17	15–19	16–18	multivariate
Dissolved oxygen	% sat	50	30–70	40–60	multivariate
Initial viable cell conc.	e6 cells/mL	1	0.5–1.5	0.8–1.2	univariate
Basal medium concentration	X	1.2	0.8–1.6	1–1.4	multivariate
Nutrient feed-1 volume	% WV	12	9–15	10–14	multivariate

Table 6.2: Design-of-experiments factor legend (coded A–E).

Code	Factor	Unit	Studied in
A	Culture pH	pH	screening + RSM
B	Culture temperature	°C	screening + RSM
C	Dissolved CO ₂ (pCO ₂)	mmHg	screening + RSM
D	Osmolality	mOsm	screening
E	Culture duration	days	screening + RSM

6.2 Screening design

Factor screening will use a resolution-V two-level fractional factorial in the five multivariate factors (A–E) with three center-point replicates, 19 runs in total. At resolution V, main effects are unconfounded with one another and with two-factor interactions, permitting clean estimation of

both. Five responses will be recorded for each run (afucosylation, galactosylation, high mannose, acidic charge variants and aggregate). The planned coded design is given in Appendix A.

6.3 Response-surface design

The dominant factors carried forward from screening (anticipated to be culture pH, temperature, culture duration and dissolved CO₂) will be studied in a face-centred central-composite design with four center-point replicates, 28 runs in total, enabling estimation of curvature in addition to main effects and interactions. The planned coded design is given in Appendix B.

6.4 Univariate assessment

Initial viable cell concentration will be evaluated one factor at a time across its characterization range for its effect on peak viable cell density, titer and culture trajectory.

7 Acceptance and decision criteria

The study is acceptable when:

1. the SDM qualification confirms representativeness of commercial scale (**SOP-1001**);
2. the fitted response-surface models are adequate — target $R^2 = 0.90$, adjusted $R^2 = 0.80$, no significant lack of fit ($p > 0.05$ against the center-point pure error), and residual diagnostics without systematic structure; and
3. a design space exists over which every in-scope CQA is satisfied simultaneously at the set-point and across each parameter's NOR.

Each parameter will then be classified per the A-Mab continuum (CMC Biotech Working Group 2009):

- a parameter whose variability significantly impacts a CQA is a **CPP**; where the impact is reliably controlled within the design space (low risk of leaving it) the parameter is a **well-controlled CPP (WC-CPP)**;
- a parameter that does not significantly impact a CQA but affects process performance or consistency is a **KPP**; otherwise it is a **GPP**;
- consistent with the post-characterization FMEA convention, a parameter that affects a CQA with Severity 8, or that carries an initial (pre-mitigation) risk priority number > 72 , is designated a CPP before mitigation by the control strategy.

The design space, classifications and PARs established here will justify the Stage 2 operating ranges and feed the drug-substance control strategy.

8 Data management and integrity

All raw data, DoE designs, analytical results and statistical analyses will be recorded and retained under the site data-integrity procedures (ALCOA+). Analyses will be performed with version-controlled scripts and reviewed by a second analyst; designs, models and derived design spaces will be archived with the report. Any deviation from this plan will be documented and assessed for impact in PCR-003.

9 Roles and responsibilities

Table 9.1: Roles and responsibilities.

Function	Responsibility
Process Development / MSAT	Author the plan; design, execute and analyze the DoE; author PCR-003
Analytical Development	Perform released and in-process testing under the referenced AMV methods
Manufacturing Science	Qualify the scale-down model; confirm commercial-scale representativeness
Statistics	Review the design, the statistical analysis and the design-space definition
Quality Assurance	Approve the plan and the report; confirm parameter classifications

10 Deliverables and schedule

The deliverable of this plan is the Process Characterization Report (**PCR-003**), reporting the executed designs, the significant effects, the design space, the parameter classification and the CQA outcomes. PCR-003 rolls up into the Process Characterization Master Report (**PCMR-001**) and provides the parameter risk basis for the post-characterization process risk assessment. Execution follows SDM qualification and precedes the Stage 2 PPQ campaign.

11 Risks and assumptions

The plan assumes the SDM is representative of commercial scale (confirmed by qualification) and that the platform analytical methods are validated and in control. The principal risk is that absolute CQA values or the harvest impurity load shift at commercial scale; this is mitigated by defining the design space on the qualified SDM, confirming it at scale in Stage 2, and carrying the WC-CPP ranges forward with in-process monitoring.

12 Approvals

Approval of this plan authorizes execution of the characterization described herein. Signatures are recorded in Table [1](#).

References

- CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.
- International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.
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- International Society for Pharmaceutical Engineering. 2023. *Good Practice Guide: Practical Implementation of the Lifecycle Approach to Process Validation*. ISPE.
- Parenteral Drug Association. 2013. *Technical Report No. 60: Process Validation — a Lifecycle Approach*. PDA.
- U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.

Appendix A — Planned screening design

Coded resolution-V fractional-factorial design (factors A–E per Table 6.2; coded levels $-1/0/+1$). Responses will be recorded at execution and reported in PCR-003.

Table 12.1: Appendix A — planned screening design (coded factor settings).

Run	Type	A	B	C	D	E
1	factorial	-1	-1	-1	-1	1
2	factorial	-1	-1	-1	1	-1
3	factorial	-1	-1	1	-1	-1
4	factorial	-1	-1	1	1	1
5	factorial	-1	1	-1	-1	-1
6	factorial	-1	1	-1	1	1
7	factorial	-1	1	1	-1	1
8	factorial	-1	1	1	1	-1
9	factorial	1	-1	-1	-1	-1
10	factorial	1	-1	-1	1	1
11	factorial	1	-1	1	-1	1
12	factorial	1	-1	1	1	-1
13	factorial	1	1	-1	-1	1
14	factorial	1	1	-1	1	-1
15	factorial	1	1	1	-1	-1
16	factorial	1	1	1	1	1
17	center	0	0	0	0	0
18	center	0	0	0	0	0
19	center	0	0	0	0	0

Appendix B — Planned response-surface design

Coded face-centred central-composite design (factors A, B, C, E; coded levels $-1/0/+1$).

Table 12.2: Appendix B — planned response-surface design (coded factor settings).

Run	Type	A	B	C	E
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	1	-1
3	factorial	-1	-1	-1	1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	1	-1
7	factorial	-1	1	-1	1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	1	-1
11	factorial	1	-1	-1	1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	1	-1
15	factorial	1	1	-1	1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	0	-1
22	axial	0	0	0	1
23	axial	0	0	-1	0
24	axial	0	0	1	0
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0